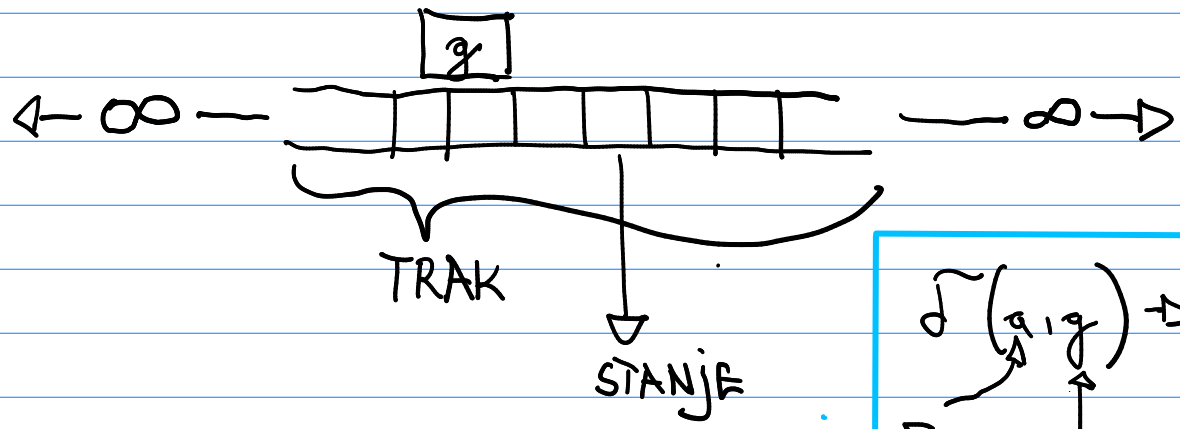
TURINGOV STROJ



$$\delta(q, g) \to (q', b, L)$$

$\Sigma \subseteq \Gamma$ $\quad$ Q

VAJA 1

SODA BINARNA ŠT.

$L\{ w0 \mid w \in \{0,1\}^* \}$

PREHOD V STANJE

KAM GREŠ

KAJ NAPIŠEŠ

$$\delta(q_0, 0) \to (q_0, 0, R)$$
$$\delta(q_0, 1) \to (q_0, 1, R)$$
$$\delta(q_1, B) \to (q_1, B, L)$$
$$\delta(q_1, 0) \to (q_F, 0, L)$$

KDAJ JE KONEC?

1. δ NI DEF

2. STANJE $\in F$

ČE SMO V STANJU q_1

IN VIDIMO 0

TRENUTNI OPIS

AKA SNAPSHOT

$\alpha \; q \; a \; \beta$

ZA GLAVO

POD GLAVO

STANJE

PRED GLAVO

$\boxed{W = 100}$

$B \; q_0 \; 100B$

$1 \; q_0 \; 00B$

$10 \; q_0 \; 0B$

$100 \; q_0 \; B$

$10 \; q_1 \; 0$

$1 \; q_F \; 00$

STANJE

POWAH OF TM

PRISTEVANJE v b

0101110 + 1

0101111

GREM DO KONCA

ZADI JE 0; DAŠ 1 IN KONC

ČENE NESEŠ 1 DO 0-ke ali B-ja

$\delta(q_0, 0) \to (q_0, 0, R)$

$\delta(q_0, 1) \to (q_0, 1, R)$

$\delta(q_0, \#) \to (q_0, \#, L)$

$\delta(q_1, 0) \to (q_1, 1, R)$

$\delta(q_1, 1) \to (q_1, 0, L)$

$\delta(q_1, B) \to (q_2, 1, L)$

2. VAJA

$L = \{a^n b^n c^n \mid n \geq 0\}$

$\overbrace{a \quad a \quad a}^{a} \overbrace{b \quad b \quad b}^{b} \overbrace{c \quad c \quad c}^{c}$

$\delta(q_0, a) \to \delta(q_1, X, R)$

$\delta(q_1, a) \to \delta(q_1, a, R)$

$\delta(q_1, Y) \to \delta(q_1, Y, R)$

$\delta(q_1, b) \to \delta(q_2, Y, R)$

$\delta(q_2, b) \to \delta(q_2, b, R)$

$\delta(q_2, Z) \to \delta(q_2, Z, R)$

$\delta(q_2, c) \to \delta(q_3, Z, L)$

ZA VSAK a MARK-AŠ 1 b IN 1 c.

NEKA KOL STANJA

POL GREŠ NAZAJ

POL PREČEKERAŠ ČE SI ZAMARKIRAL VSE IN SI PRŠU DO "B"

MNOŽENJE

$L = \{ a^n b^m c^{n \times m} \}$

A SE DA? FUCK YEA! ... FUN ...

a b c

$\delta(q_0, a) \to (q_1, X, R)$
$\delta(q_1, a) \to (q_1, a, R)$
$\delta(q_1, b) \to (q_2, Y, R)$
$\delta(q_2, b) \to (q_2, b, R)$
$\delta(q_2, Z) \to (q_3, Z, L)$
$\delta(q_3, Z) \to (q_3, Z, L)$
$\delta(q_3, b) \to (q_3, b, L)$
$\delta(q_3, Y) \to (q_2, Y, R)$
$\delta(q_2, Z) \to (q_4, Z, L)$
$\delta(q_4, Y) \to (q_4, b, L)$
$\delta(q_4, a) \to (q_4, a, L)$
$\delta(q_4, X) \to (q_0, X, R)$
$\delta(q_0, b) \to (q_5, b, R)$
$\delta(q_5, b) \to (q_5, b, R)$
$\delta(q_5, Z) \to (q_6, Z, R)$
$\delta(q_6, Z) \to (q_6, Z, R)$
$\delta(q_6, B) \to (q_{f1}, B, L)$

LOOP
FOR j IN (a)

LOOP
FOR i IN (b)

DAŠ VSE Y NAZAJ NA b

GREŠ NA ZAČETEK

PREČEKERAŠ

Tj DR

q_1 – PRENOS A-JEV
q_2 – OZNAČI B / PRENAŠA B
q_3 – GRE NAZAJ

TO GOVNO MA MOGOČE BUG :) $\boxed{\text{FML}}$

IDEJA

a a b b c c c c
x a b b c c c c
x a y b z c c c
x a y y z z c c
x a b b z z c c
x x y b z z z c
x x y y z z z z
x x b b z z z z

TIP! NAPIŠI KONCEPT/IDEJO STROJA NA IZPITU

LOGARITEM BRT !

a	BINARNO
$a\,a$	1000
$a\,a\,a\,a$	$1\,0^*$ → izi
a^B	

1) UGOTOVIŠ A JE NIZ SOD

2) DELIŠ Z 2

$$a\,a\,a\,a$$
$$\downarrow$$
$$a\,X\,a\,X$$
$$\downarrow$$
$$a\,X\,X\,X$$

USAKGA DRUGGA VN FUKNEŠ = /2

$$\left.\begin{array}{l}(g_1, a) \to (g_2, a, L) \\ \quad a \text{ OR } b \qquad a \text{ OR } b \\ (g_1, b) \to (g_2, b, L)\end{array}\right\} \boxed{\text{SHORTCUT} \quad \begin{pmatrix} a \\ g_1, b \end{pmatrix} \to \begin{pmatrix} a \\ g_2, b, L \end{pmatrix}}$$